

SMART CHASIS VIBRATION ANALYSIS SYSTEM

Topic: Chassis Structures and Safety Telltales

Submitted by: Sneha Talukdar

College: Institute of Engineering and Management, New Town

Course: Wipro Automotive Course, 2027

Technology Used: Python, Pandas, NumPy, Matplotlib, Scikit-learn, Google Colab

Machine Learning Model: Random Forest Classifier

1. INTRODUCTION:

The chassis is a major structural component of a vehicle that supports key systems and experiences various loads and vibrations during operation. Continuous or abnormal vibrations may indicate issues such as- loose joints, structural irregularities, or mechanical faults. Therefore, vibration monitoring can help in the early identification of abnormal vehicle behaviour.

Accelerometers can measure vibration and acceleration at different points of the vehicle. These measurements can be processed to obtain parameters such as- acceleration level, vibration frequency, and RMS (Root Mean Square) vibration.

In this project, a Smart Chassis Vibration Analysis System is developed as a simulated machine-learning prototype. Since, physical accelerometer hardware and real vehicle data were unavailable, synthetic data representing Normal, Warning, and Critical conditions was generated. The data was analyzed using Python and used to train a Random Forest classification model.

The prototype demonstrates how sensor-based vibration data can be analyzed and classified into different chassis conditions. A safety telltale

system is also implemented to indicate the predicted state as Normal, Warning, or Critical.

2. OBJECTIVES:

The main objectives are:

- To simulate accelerometer-based chassis vibration data.
- To analyze acceleration, frequency, and RMS vibration.
- To visualize vibration patterns using graphs and frequency analysis.
- To develop a machine-learning model for chassis-condition classification.
- To implement a Normal, Warning, and Critical safety telltale.
- To demonstrate the use of vibration analysis for vehicle structural-condition monitoring.

3. METHODOLOGY:

The proposed system follows a sequence of sensor-data generation, data processing, vibration analysis, machine-learning classification, and safety indication.

The overall workflow of the proposed system is:

Accelerometer Data -> Data Processing -> Vibration Analysis -> Machine Learning Model -> Safety Telltale

The system considers three vibration conditions:

- Normal – Low-level vibration representing normal operating behaviour.
- Warning – Increased vibration representing an abnormal condition that may require attention.
- Critical – High vibration representing a potentially severe abnormal condition requiring immediate attention.

Since, physical accelerometer hardware and real vehicle measurements were not available, synthetic accelerometer-based vibration data was

generated using Python. The dataset contains acceleration, frequency, and RMS vibration values along with their corresponding condition labels.

The generated data was processed using Pandas and NumPy and analyzed using numerical calculations and visualization. The selected features were then used to train a Random Forest Classifier using Scikit-learn.

The trained model predicts the chassis condition as Normal, Warning, or Critical. The predicted condition is then displayed through a safety telltale system.

4. DATASET AND DATA GENERATION:

Since, real-time accelerometer hardware and vehicle vibration data were not available, a synthetic dataset was generated using Python.

The dataset represents three chassis conditions: Normal, Warning, and Critical. The main parameters used were:

- Acceleration
- Frequency
- RMS Vibration

The data was generated with different vibration characteristics for each condition so that the machine-learning model could learn to distinguish between normal and abnormal chassis behaviour.

The final dataset was processed using Pandas and NumPy and used for training and testing the Random Forest Classification Model.

5. DATA PREPROCESSING:

The generated dataset was loaded and checked by using Pandas. The data was organized into input features and the target condition.

The three input features used were:

- Acceleration

- Frequency
- RMS Vibration

The target variable contained three classes: Normal, Warning, and Critical.

The dataset was then divided into training and testing sets. The training data was used to build the Random Forest model, while the testing data was used to evaluate its performance on unseen data.

6. VIBRATION ANALYSIS:

The generated vibration data was analyzed using acceleration, frequency, and RMS vibration to identify differences between the three chassis conditions.

Graphs were created by using Matplotlib to visualize the vibration patterns. RMS vibration was used to represent the overall intensity of the vibration signal, while frequency helped identify the vibration characteristics of different conditions.

The analysis provides a basis for distinguishing Normal, Warning, and Critical chassis conditions before applying machine-learning classification.

7. MACHINE LEARNING MODEL:

A Random Forest Classifier was used to classify the chassis condition into Normal, Warning, and Critical categories.

The model was trained using the three selected features: Acceleration, Frequency, and RMS Vibration. Random Forest was chosen because it can handle multiple input features and is suitable for classification problems.

The trained model was evaluated using the test dataset. Feature importance was also calculated to understand the contribution of each input parameter to the classification.

The obtained feature importance values were:

- Frequency: 0.3596
- Acceleration: 0.3205
- RMS Vibration: 0.3199

Among the three features, frequency showed the highest contribution to the model's classification.

8. RESULTS AND PERFORMANCE:

The Random Forest Classifier successfully classified the chassis vibration data into Normal, Warning, and Critical conditions.

The model achieved 100% accuracy on the test dataset. The classification report showed a precision, recall, and F1-score of 1.00 for all three classes.

The confusion matrix also showed that all test samples were correctly classified into their respective categories, with no misclassification between Normal, Warning, and Critical conditions.

The results demonstrate that the trained model was able to effectively distinguish between the simulated chassis vibration conditions.

9. SAFETY TELLTALE, CONCLUSION, FUTURE SCOPE:

A safety telltale system was implemented to display the predicted chassis condition. Based on the machine-learning output, the system indicated Normal, Warning, or Critical status, helping the user understand the current vibration condition.

The project demonstrates how accelerometer-based vibration parameters such as- acceleration, frequency, and RMS vibration can be analyzed using machine learning for chassis-condition monitoring. The Random Forest Classifier achieved 100% accuracy on the generated text dataset and correctly classified the three simulated conditions.

In future, the system can be improved by using real-time accelerometer sensors and real vehicle vibration data. Additional features, larger

datasets, and real-time accelerometer sensors and real vehicle vibration data. Additional features, larger datasets, and real-time monitoring could further improve the practical applicability of the system. The safety telltale could also be integrated with a vehicle dashboard or warning system.